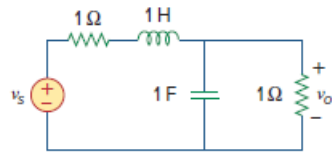


Problem

If v_s , in the circuit of Fig. 17.71 is the same as function $f_2(t)$ in Fig. 17.56(b), determine the dc component and the first three nonzero harmonics of $v_o(t)$.

Figure 17.71:



Step-by-step solution

Step 1 of 9

Refer to Figure 17.56 (b) in the textbook for the input voltage.

The expression for the input voltage from the waveform is,

$$v_s(t) = \begin{cases} 1 \text{ V} & 0 < t < 1 \\ 2 \text{ V} & 1 < t < 2 \\ 1 \text{ V} & 2 < t < 3 \end{cases}$$

The time period is,

$$T = 3 \text{ s}$$

Calculate the angular frequency.

$$\begin{aligned} \omega_o &= \frac{2\pi}{T} \\ &= \frac{2\pi}{3} \text{ rad/s} \end{aligned}$$

Calculate the value of Fourier coefficient, a_0 .

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T v_s(t) dt \\ &= \frac{1}{3} \left[\int_0^1 (1) dt + \int_1^2 (2) dt + \int_2^3 (1) dt \right] \\ &= \frac{1}{3} [1 + 2 + 1] \\ &= \frac{4}{3} \end{aligned}$$

Step 2 of 9

Calculate the value of Fourier coefficient, a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T v_s(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{3} \left[\int_0^1 (1) \cos(n\omega_0 t) dt + \int_1^2 (2) \cos(n\omega_0 t) dt + \int_2^3 (1) \cos(n\omega_0 t) dt \right] \\
 &= \frac{2}{3} \left[\left[\frac{\sin(n\omega_0 t)}{n\omega_0} \right]_0^1 + 2 \left[\frac{\sin(n\omega_0 t)}{n\omega_0} \right]_1^2 + \left[\frac{\sin(n\omega_0 t)}{n\omega_0} \right]_2^3 \right] \\
 &= \frac{2}{3n\omega_0} [\sin(n\omega_0) + 2\sin(2n\omega_0) - 2\sin(n\omega_0) + \sin(3n\omega_0) - \sin(2n\omega_0)] \\
 &= \frac{2}{3n\omega_0} [\sin(2n\omega_0) - \sin(n\omega_0) + \sin(3n\omega_0)] \\
 &= \frac{2}{3n \times \frac{2\pi}{3}} \left[\sin\left(2n \times \frac{2\pi}{3}\right) - \sin\left(n \times \frac{2\pi}{3}\right) + \sin\left(3n \times \frac{2\pi}{3}\right) \right] \\
 &= \frac{1}{n\pi} \left[\sin\left(\frac{4n\pi}{3}\right) - \sin\left(\frac{2n\pi}{3}\right) + \sin(2n\pi) \right] \\
 &= \frac{1}{n\pi} \left[\sin\left(2n\pi - \frac{2n\pi}{3}\right) - \sin\left(\frac{2n\pi}{3}\right) + 0 \right] \\
 &= \frac{1}{n\pi} \left[-\sin\left(\frac{2n\pi}{3}\right) - \sin\left(\frac{2n\pi}{3}\right) \right] \\
 &= -\frac{2}{n\pi} \sin\left(\frac{2n\pi}{3}\right)
 \end{aligned}$$

Step 3 of 9

Calculate the value of Fourier coefficient, b_n .

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T v_s(t) \sin(n\omega_0 t) dt \\
 &= \frac{2}{3} \left[\int_0^1 (1) \sin(n\omega_0 t) dt + \int_1^2 (2) \sin(n\omega_0 t) dt + \int_2^3 (1) \sin(n\omega_0 t) dt \right] \\
 &= -\frac{2}{3} \left[\left[\frac{\cos(n\omega_0 t)}{n\omega_0} \right]_0^1 + 2 \left[\frac{\cos(n\omega_0 t)}{n\omega_0} \right]_1^2 + \left[\frac{\cos(n\omega_0 t)}{n\omega_0} \right]_2^3 \right] \\
 &= -\frac{2}{3n\omega_0} [\cos(n\omega_0) - 1 + 2\cos(2n\omega_0) - 2\cos(n\omega_0) + \cos(3n\omega_0) - \cos(2n\omega_0)] \\
 &= -\frac{2}{3n\omega_0} [\cos(2n\omega_0) - 1 - \cos(n\omega_0) + \cos(3n\omega_0)] \\
 &= -\frac{2}{3n \times \frac{2\pi}{3}} \left[\cos\left(2n \times \frac{2\pi}{3}\right) - 1 - \cos\left(n \times \frac{2\pi}{3}\right) + \cos\left(3n \times \frac{2\pi}{3}\right) \right] \\
 &= -\frac{1}{n\pi} \left[\cos\left(\frac{4n\pi}{3}\right) - 1 - \cos\left(\frac{2n\pi}{3}\right) + \cos(2n\pi) \right] \\
 &= -\frac{1}{n\pi} \left[\cos\left(2n\pi - \frac{2n\pi}{3}\right) - 1 - \cos\left(\frac{2n\pi}{3}\right) + 1 \right] \\
 &= \frac{1}{n\pi} \left[\cos\left(\frac{2n\pi}{3}\right) - 1 - \cos\left(\frac{2n\pi}{3}\right) + 1 \right] \\
 &= 0
 \end{aligned}$$

Thus, the expression for the input source voltage is,

$$\begin{aligned}
 v_s(t) &= a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \\
 &= \frac{4}{3} - \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\left(\frac{2n\pi}{3}\right) \cos\left(\frac{2n\pi t}{3}\right) + (0) \sin\left(\frac{2n\pi t}{3}\right) \\
 &= \frac{4}{3} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{2n\pi}{3}\right) \cos\left(\frac{2n\pi t}{3}\right)
 \end{aligned}$$

Refer to Figure 17.71 in the textbook for the circuit diagram.

The angular frequency for the phasor circuit is,

$$\begin{aligned}
 \omega_n &= n\omega_0 \\
 &= \frac{2n\pi}{3}
 \end{aligned}$$

Draw the circuit in frequency domain.

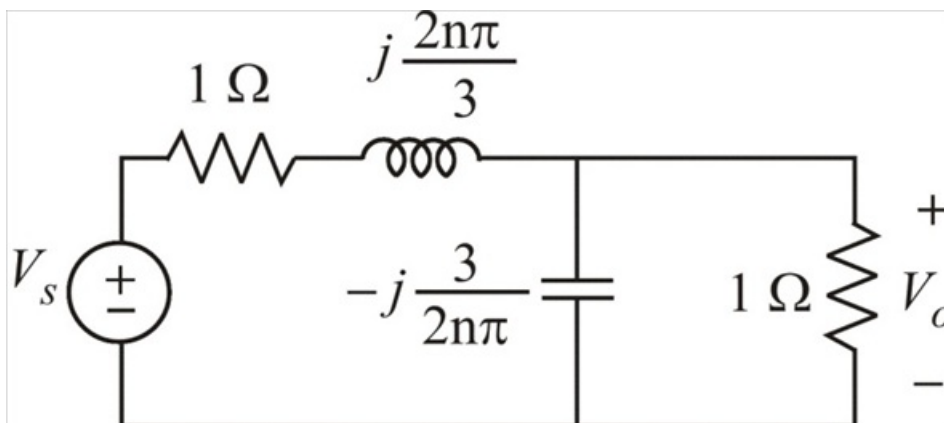


Figure 1

Calculate the impedance of the parallel capacitor and resistor.

$$\begin{aligned} Z &= (1) \parallel \left(\frac{-j3}{2n\pi} \right) \\ &= \frac{(1) \left(\frac{-j3}{2n\pi} \right)}{1 - \frac{j3}{2n\pi}} \\ &= \frac{-j3}{2n\pi - j3} \end{aligned}$$

Apply voltage division to calculate the output voltage.

$$\begin{aligned} \mathbf{V}_o &= \left(\frac{Z}{Z + 1 + j \frac{2n\pi}{3}} \right) \mathbf{V}_s \\ &= \left(\frac{3Z}{3Z + 3 + j2n\pi} \right) \mathbf{V}_s \\ &= \left[\frac{3 \left(\frac{-j3}{2n\pi - j3} \right)}{3 \left(\frac{-j3}{2n\pi - j3} \right) + 3 + j2n\pi} \right] \mathbf{V}_s \\ &= \left[\frac{-j9}{-j9 + 3(2n\pi - j3) + j2n\pi(2n\pi - j3)} \right] \mathbf{V}_s \\ &= \left(\frac{-j9}{-j9 + 6n\pi - j9 + j4n^2\pi^2 + 6n\pi} \right) \mathbf{V}_s \\ &= \left[\frac{-j9}{12n\pi + j(4n^2\pi^2 - 18)} \right] \mathbf{V}_s \end{aligned}$$

For dc component, $\omega_n = 0$ or $n = 0$.

Calculate the dc component of the output voltage.

$$\begin{aligned} V_o &= \left| \frac{-j9}{12(0)\pi + j(4(0)^2\pi^2 - 18)} \right| V_s \\ &= \left| \frac{9}{18} \right| V_s \\ &= \frac{1}{2} \left(\frac{4}{3} \right) \\ &= \frac{2}{3} \end{aligned}$$

Calculate the output voltage at the frequency $\omega_n = \frac{2n\pi}{3}$

$$\begin{aligned} v_o &= \left[\frac{-j9}{12n\pi + j(4n^2\pi^2 - 18)} \right] v_s \\ &= \sum_{n=1}^{\infty} \frac{-2}{n\pi} \left[\frac{-j9}{12n\pi + j(4n^2\pi^2 - 18)} \right] \sin\left(\frac{2n\pi}{3}\right) \cos\left(\frac{2n\pi t}{3}\right) \\ &= \sum_{n=1}^{\infty} \frac{1}{n\pi} \left[\frac{j18}{3 \left[4n\pi + j\left(\frac{4n^2\pi^2}{3} - 6\right) \right]} \right] \sin\left(\frac{2n\pi}{3}\right) \cos\left(\frac{2n\pi t}{3}\right) \\ &= \sum_{n=1}^{\infty} \left[\frac{\frac{j6}{n\pi} \sin\left(\frac{2n\pi}{3}\right)}{4n\pi + j\left(\frac{4n^2\pi^2}{3} - 6\right)} \right] \cos\left(\frac{2n\pi t}{3}\right) \end{aligned}$$

Calculate the magnitude of the Fourier coefficient of output.

$$A_n = \frac{\frac{6}{n\pi} \sin\left(\frac{2n\pi}{3}\right)}{\sqrt{(4n\pi)^2 + \left(\frac{4n^2\pi^2}{3} - 6\right)^2}}$$

Calculate the phase of the Fourier coefficient of output.

$$\begin{aligned} \theta_n &= \frac{\pi}{2} - \tan^{-1} \left(\frac{\frac{4n^2\pi^2}{3} - 6}{4n\pi} \right) \\ &= \frac{\pi}{2} - \tan^{-1} \left(\frac{n\pi}{3} - \frac{3}{2n\pi} \right) \end{aligned}$$

Step 6 of 9

The overall expression for the output voltage is,

$$v_o = \frac{2}{3} + \sum_{n=1}^{\infty} A_n \cos\left(\frac{2n\pi t}{3} + \theta_n\right)$$

where,

$$\begin{aligned} A_n &= \frac{\frac{6}{n\pi} \sin\left(\frac{2n\pi}{3}\right)}{\sqrt{(4n\pi)^2 + \left(\frac{4n^2\pi^2}{3} - 6\right)^2}} \\ \theta_n &= \frac{\pi}{2} - \tan^{-1} \left(\frac{n\pi}{3} - \frac{3}{2n\pi} \right) \end{aligned}$$

Step 7 of 9

Calculate the magnitude and phase of first harmonic of the output voltage.

$$A_1 = \frac{\frac{6}{\pi} \sin\left(\frac{2\pi}{3}\right)}{\sqrt{(4\pi)^2 + \left(\frac{4\pi^2}{3} - 6\right)^2}}$$

$$= 0.114$$

$$\theta_1 = \frac{\pi}{2} - \tan^{-1}\left(\frac{\pi}{3} - \frac{3}{2\pi}\right)$$

$$= 90^\circ - 29.67^\circ$$

$$= 60.33^\circ$$

Step 8 of 9

Calculate the magnitude and phase of second harmonic of the output voltage.

$$A_2 = \frac{\frac{6}{2\pi} \sin\left(\frac{4\pi}{3}\right)}{\sqrt{(8\pi)^2 + \left(\frac{16\pi^2}{3} - 6\right)^2}}$$

$$= -0.0156$$

$$\theta_2 = \frac{\pi}{2} - \tan^{-1}\left(\frac{2\pi}{3} - \frac{3}{4\pi}\right)$$

$$= 90^\circ - 61.68^\circ$$

$$= 28.32^\circ$$

Step 9 of 9

Calculate the magnitude and phase of third harmonic of the output voltage.

$$A_3 = \frac{\frac{6}{3\pi} \sin\left(\frac{6\pi}{3}\right)}{\sqrt{(12\pi)^2 + \left(\frac{36\pi^2}{3} - 6\right)^2}}$$

$$= 0$$

Calculate the magnitude and phase of fourth harmonic of the output voltage.

$$A_4 = \frac{\frac{6}{4\pi} \sin\left(\frac{8\pi}{3}\right)}{\sqrt{(16\pi)^2 + \left(\frac{64\pi^2}{3} - 6\right)^2}}$$

$$= 0.00196$$

$$\theta_4 = \frac{\pi}{2} - \tan^{-1}\left(\frac{4\pi}{3} - \frac{3}{8\pi}\right)$$

$$= 90^\circ - 76.2^\circ$$

$$= 13.8^\circ$$

Thys, the dc component of output voltage is $\frac{2}{3}$ first three nonzero harmonics of output voltage is, $0.114\angle 60.33^\circ, -0.0156\angle 28.32^\circ, 0.00196\angle 13.8^\circ$.